Introduction to Regression
Part A - kNN



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Lecture Outline

Part A: Statistical Modeling

k-Nearest Neighbors (kNN)

Part B: Model Fitness

How well does the model predict?

How do we choose from two different models?

Response and Predictor Variables

Predicting a Variable

Suppose we want to predict the value of one variable from one or more other variables.

Examples:

Predicting the number of views that a **TikTok** video will receive next week based on factors such as video length, posting date, and previous view count.



Predicting a Variable

Suppose we want to predict the value of one variable from one or more other variables.

Examples:

Forecasting which movies, a Netflix user is likely to rate highly, considering their previous movie ratings and demographic data.



Working example

The Advertising dataset contains sales data (in 1000 units) for a specific product across 200 different markets. It also includes advertising budgets in $1000 allocated to three different media channels: *TV, radio, and newspaper*, for each of those markets.

TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

Some of the figures in this presentation are taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani "

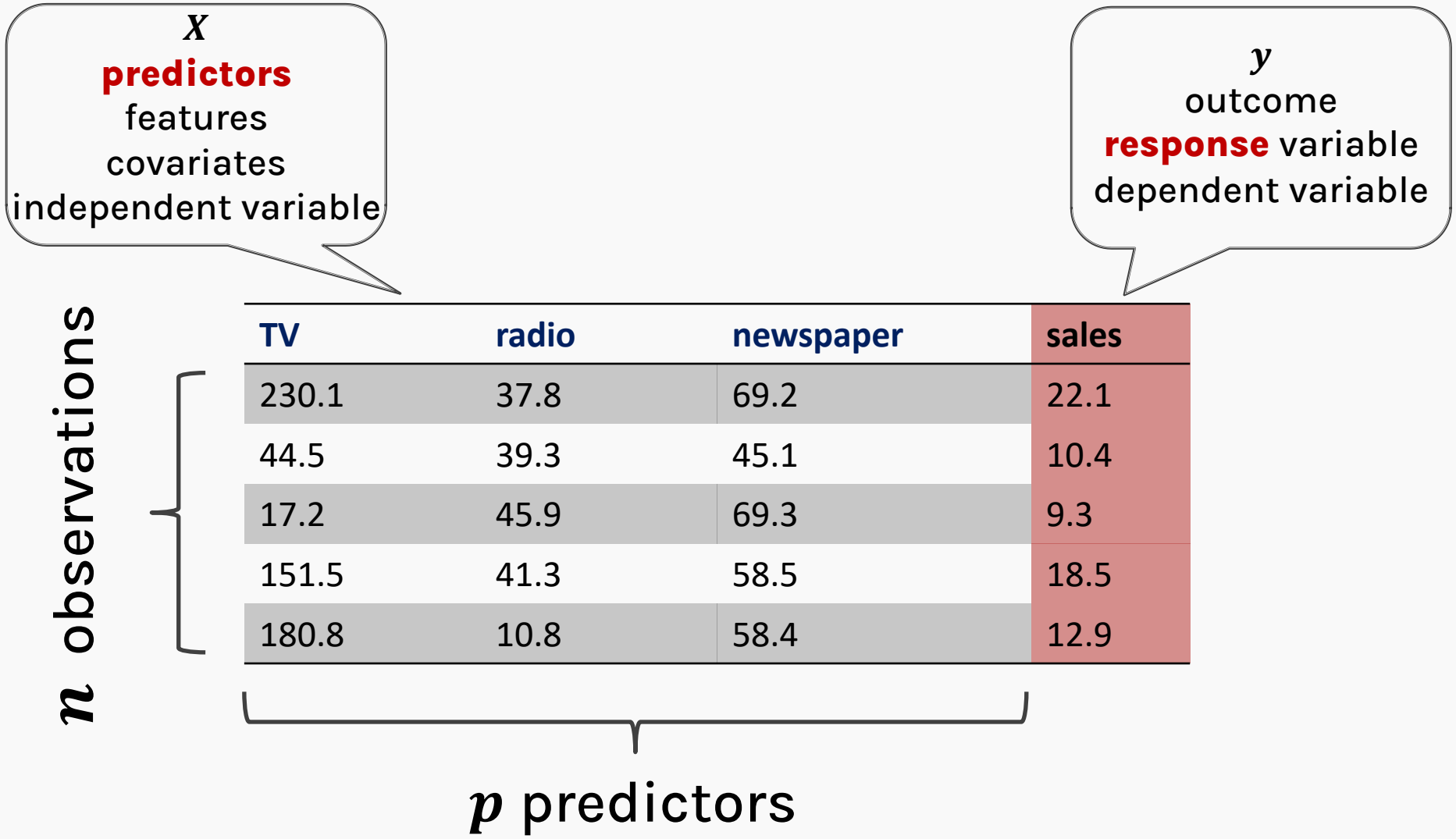
Response vs. Predictor Variables

Many of problems of this kind exhibit an asymmetry: the variable we aim to predict may be harder to measure, more consequential, or directly or indirectly influenced by other variables.

Therefore, we can classify variables into two categories:

- Variables whose values we aim to predict
- Variables used as inputs to inform our prediction

Response vs. Predictor Variables



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Response vs. Predictor Variables

$X = X_1, \dots, X_p$

$X_j = x_{1j}, \dots, x_{ij}, \dots, x_{nj}$

predictors

features

covariates

independent variable

$\boldsymbol{y} = y_1, \dots, y_i, \dots, y_n$

outcome

response variable

dependent variable

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n observations

p predictors

Response vs. Predictor Variables

This is called X: a.k.a.
The Design Matrix

n observations

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The response variable

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Capital letters mean matrices,

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Capital letters mean matrices, lower case letters mean vectors

Sklearn expects certain dimensions

```
>>> X.shape
(n,p)
```

```
>>> y.shape
(n,) OR (n,1)
```

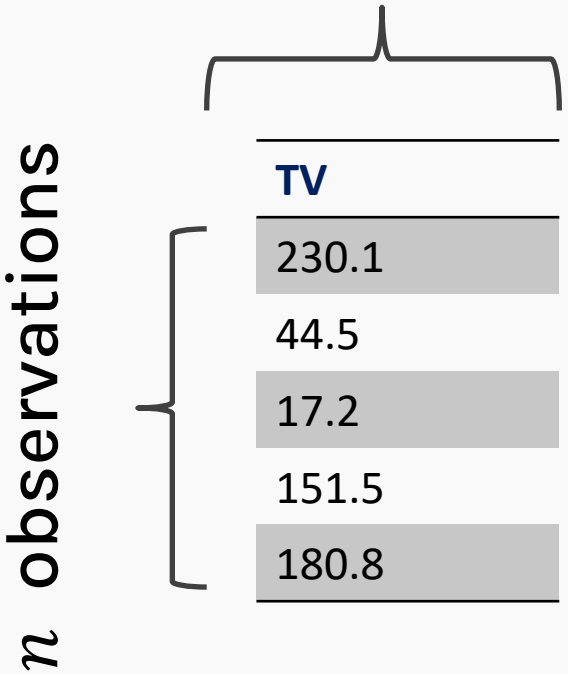
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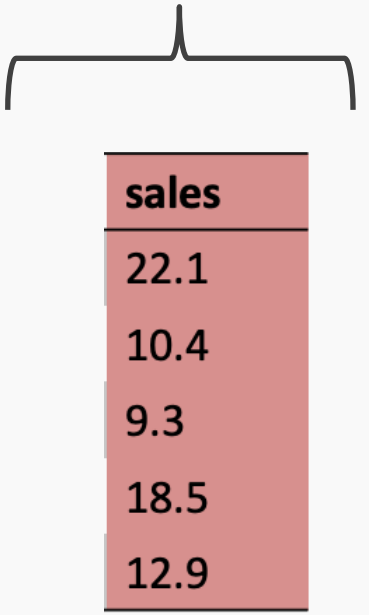
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CS109A
GAME Time

Game time: Pandas Review





df[['x']] vs df['x']

Which of the statements below is correct?

Options:

A. df[['x']] returns a pd.Series object whereas df['x'] returns a pd.DataFrame.

B. df[['x']] is invalid operation.

C. df[['x']] returns a pd.DataFrame whereas df['x'] returns a pd.Series object.

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Game time: Pandas Review





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Statistical Model

True vs. Statistical Model

We assume that the response variable, y, is related to the predictor variables, X, through an unknown function which can be generally expressed as:

$$y = f(X) + \varepsilon$$

Here, f represents the unknown function expressing an underlying rule for relating y to X. ε represents the random amount (unrelated to X) that y differs from the rule $f(X)$.

A ***statistical model*** is any algorithm used to estimate f. We denote the estimated function as $\widehat{f}$.

Prediction vs. Estimation

Inference Problems:

- The primary focus is on obtaining $\hat{f}$, which is an estimate of the true function f
- Objective: Understand the form and characteristics of $\hat{f}$.



Prediction Problems

- The specific form of $\hat{f}$ is less important than the accuracy of the predictions.
- Objective: Minimize the difference between predicted values $\hat{y}$ and observed values y.



Prediction vs. Estimation

Inference Problems

Parametric models (few interpretable parameters



Examples:

- Linear Regression
- Logistic Regression

Prediction Problems

Non-parametric / flexible models (focus on accuracy)



Examples:

- K-Nearest Neighbors
- Random Forest
- Neural Networks *(parametric but high-dimensional, used mainly for prediction)*

Prediction vs. Estimation

In this lecture, we will be talking about **K-Nearest Neighbors**

(Prediction, Non-Parametric)

Why?

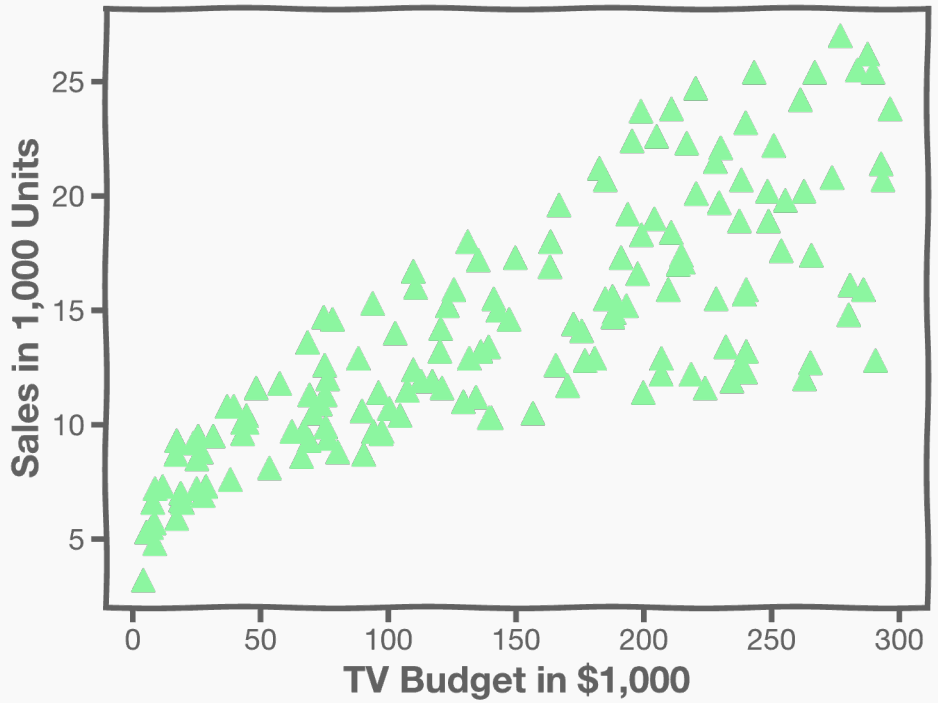


Well… because it's easier to understand.

Example: predicting sales

Motivation: Predict Sales

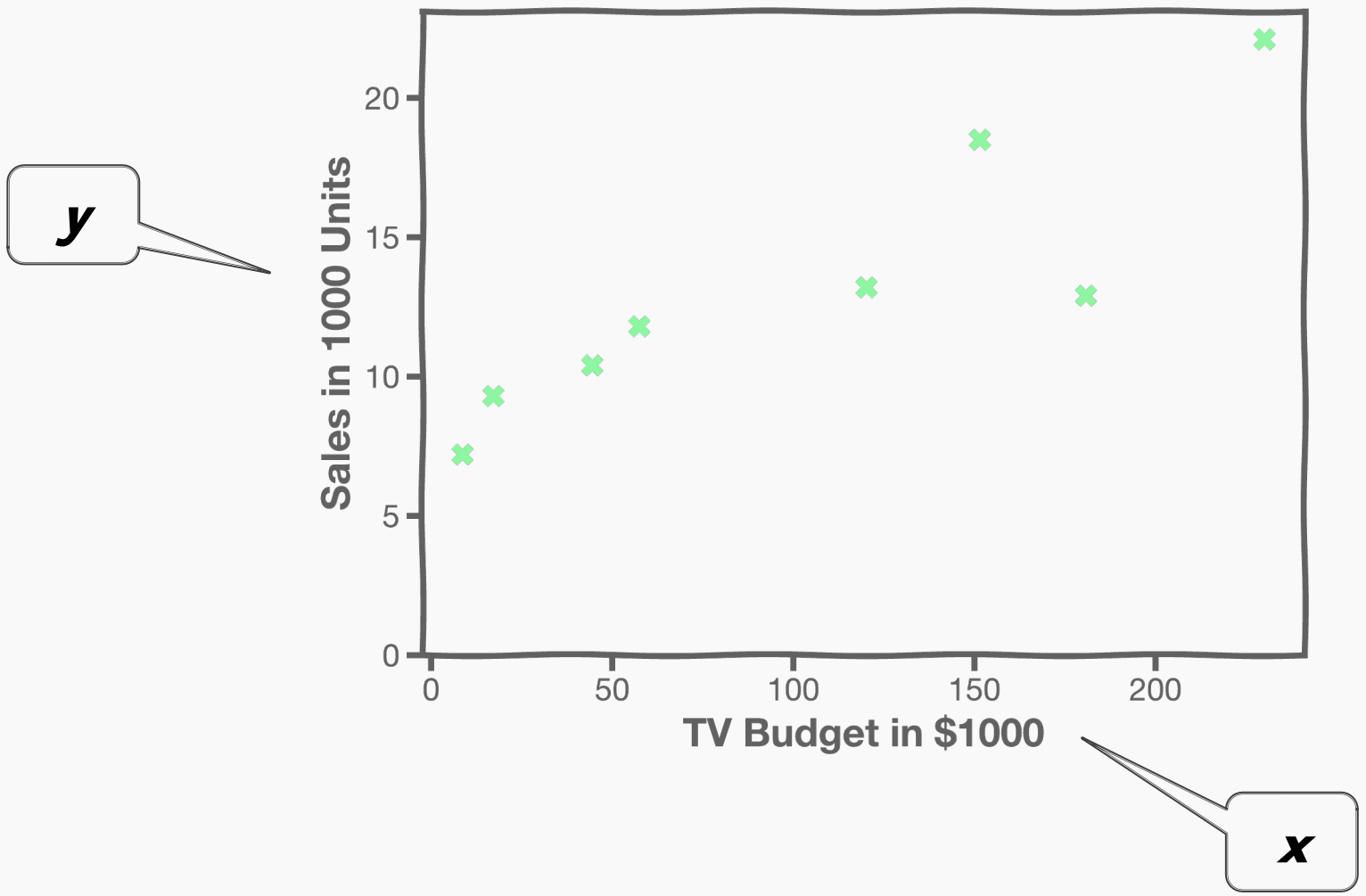
Build a model to **predict** sales based on TV budget



The response, ***y***, is the sales.

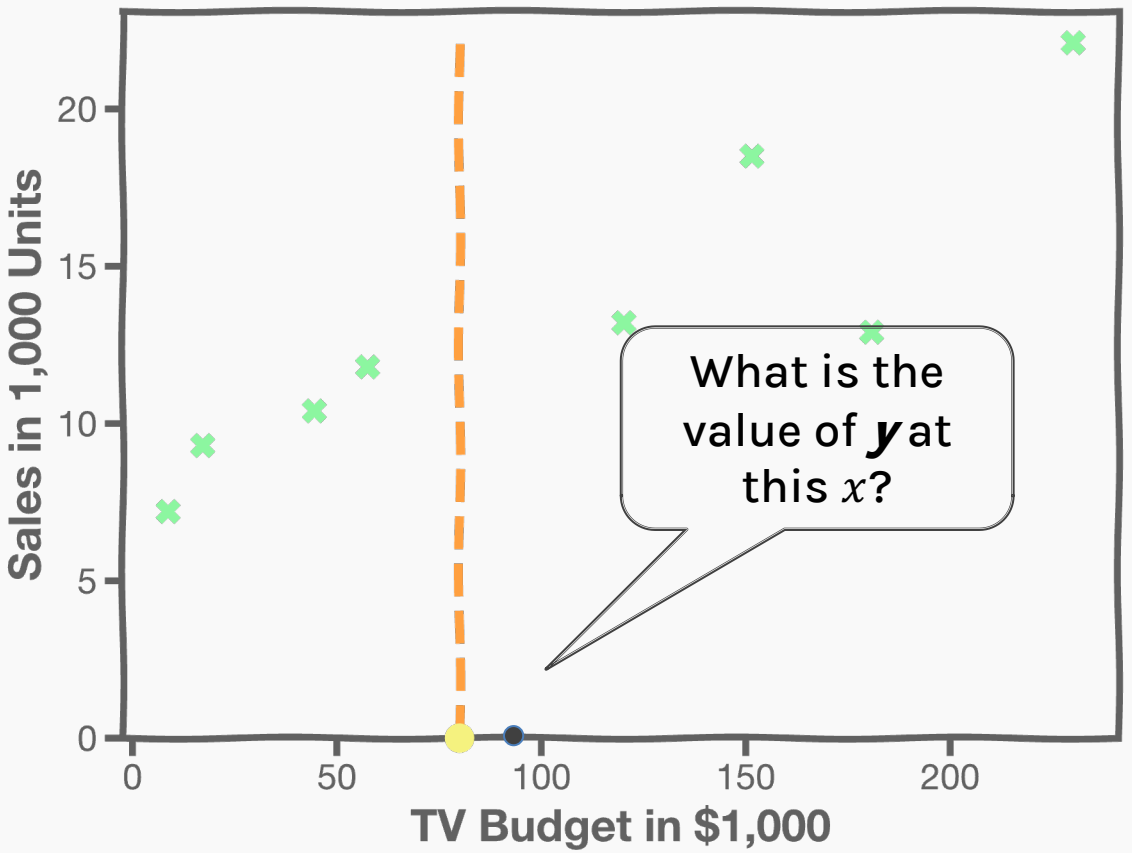
The predictor, ***x***, is TV budget.

Statistical Model



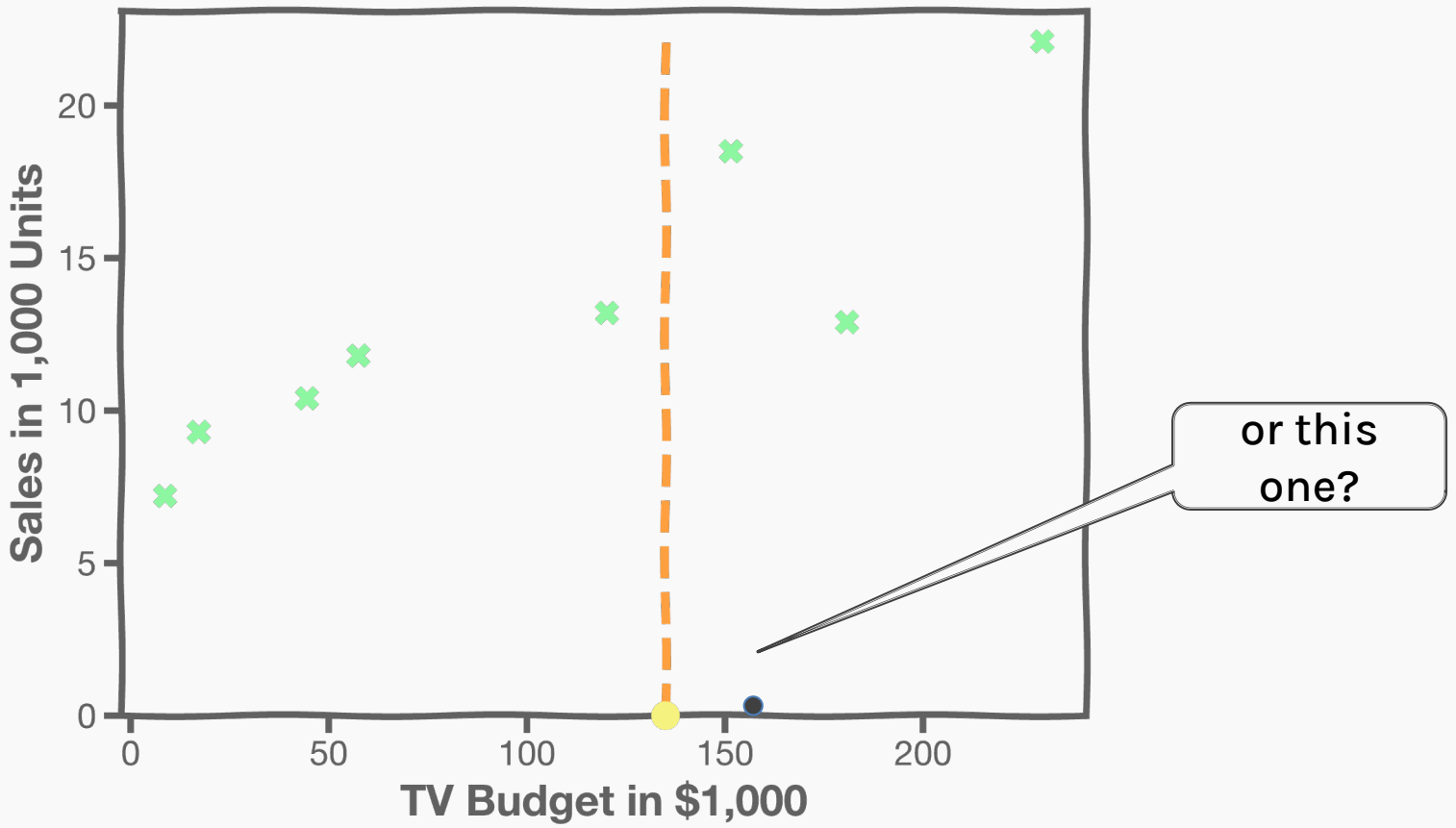
Statistical Model

How do we predict y for some x?



Statistical Model

How do we predict y for some x?



Game time: Choices for model





Which of the following 5 methods could be used to predict the value of y given x?

Options:

A. Utilize a Convolutional Neural Network (CNN).

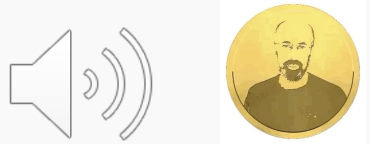
B. Use a Linear Regression Model with a slope of 3 and an intercept of 2.

C. Identify examples that closely resemble the input data point.

D. Consult a TF during office hours for the answer.

E. Calculate the average value of y from the available data points.

Game time: Choices for model



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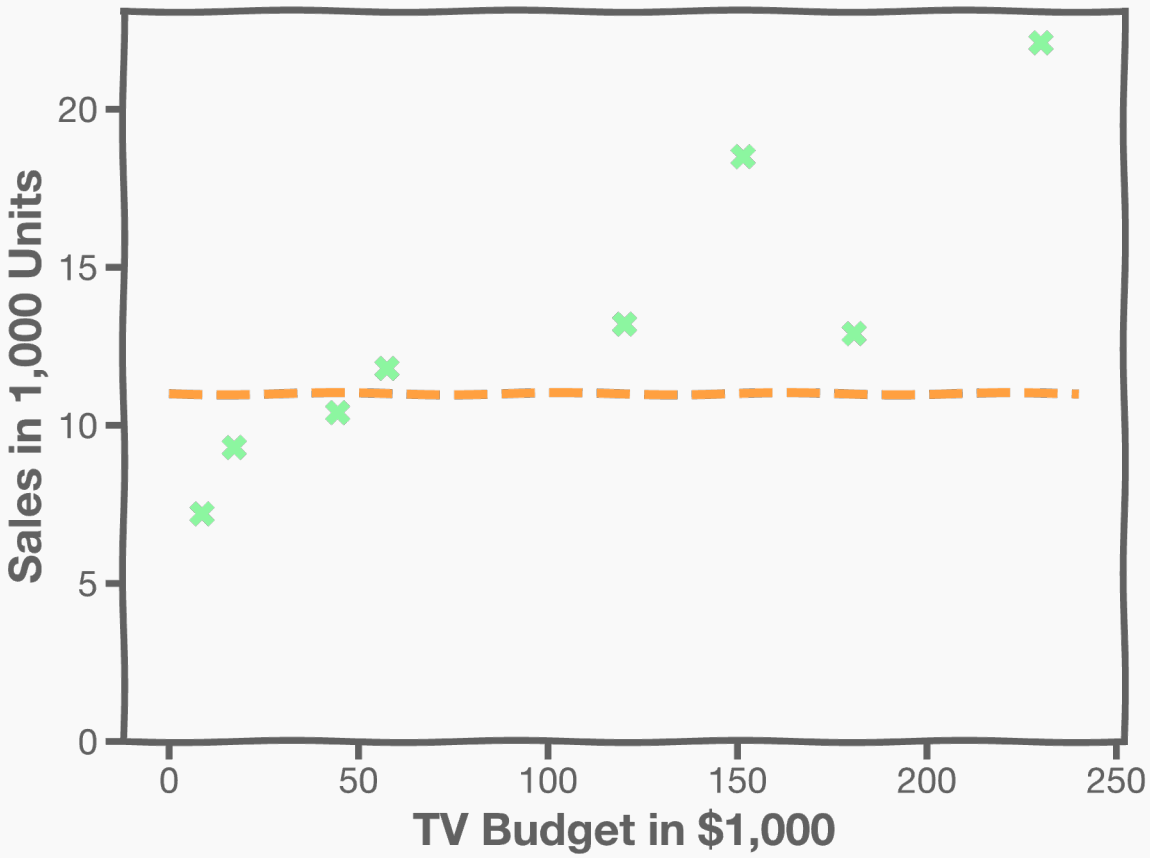
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Statistical Model

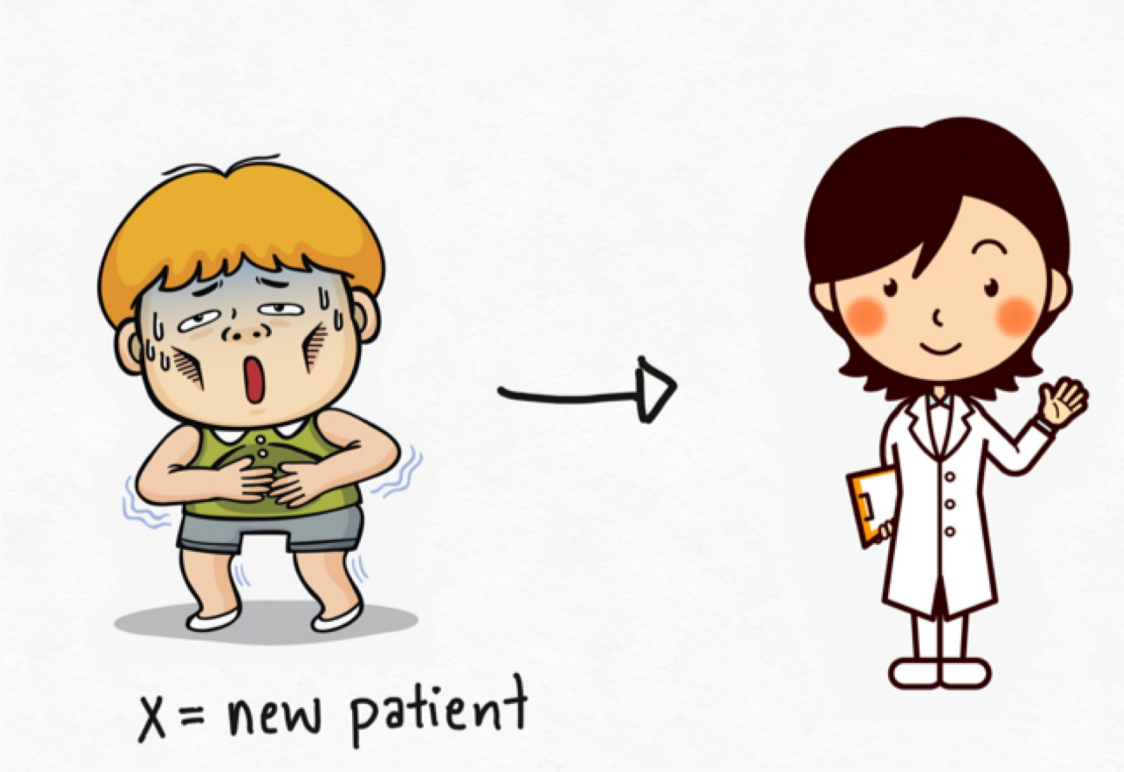
A simple idea is to take the mean of all y's: $\frac{1}{n}\sum_{i=1}^{n} y_i$



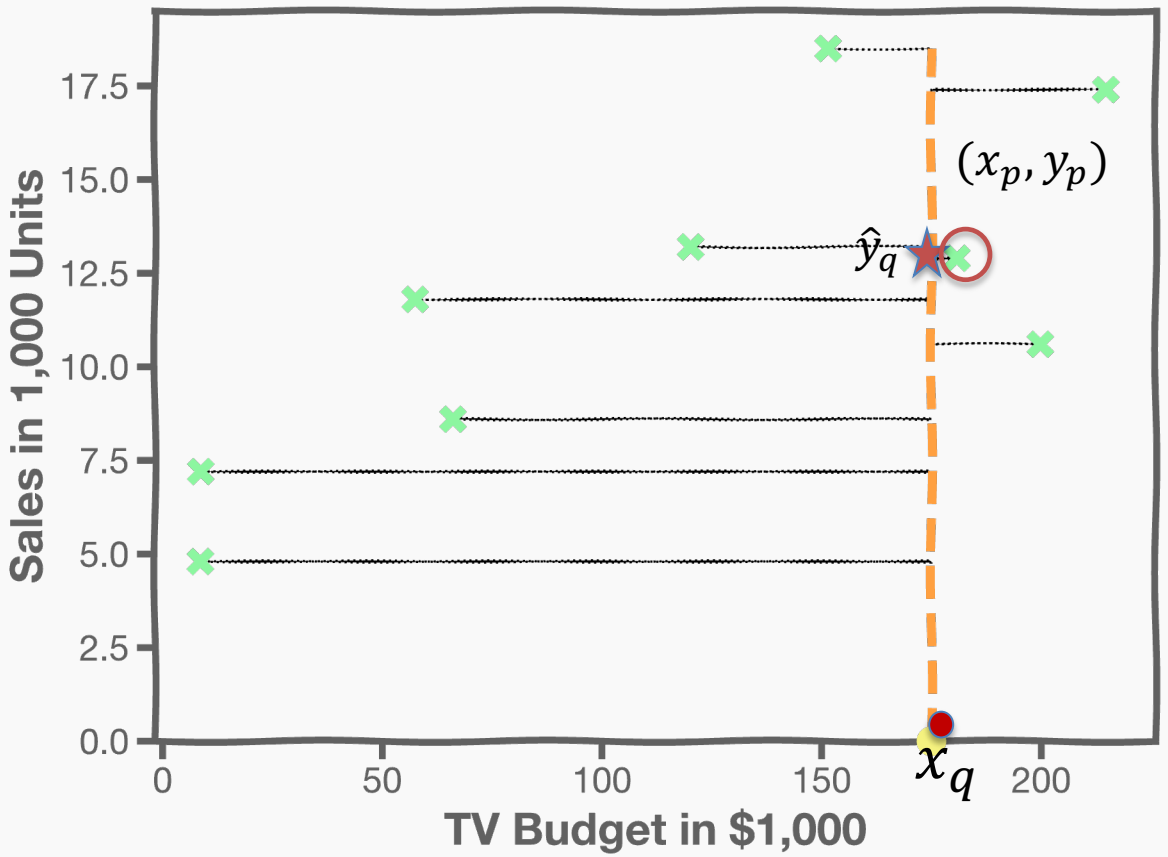
$$\frac{1}{n}\sum_{i=1}^{n} y_i$$

K-Nearest Neighbors

k-Nearest Neighbors – kNN



k-Nearest Neighbors – kNN



What is $\hat{y}_q$ at some x_q ?

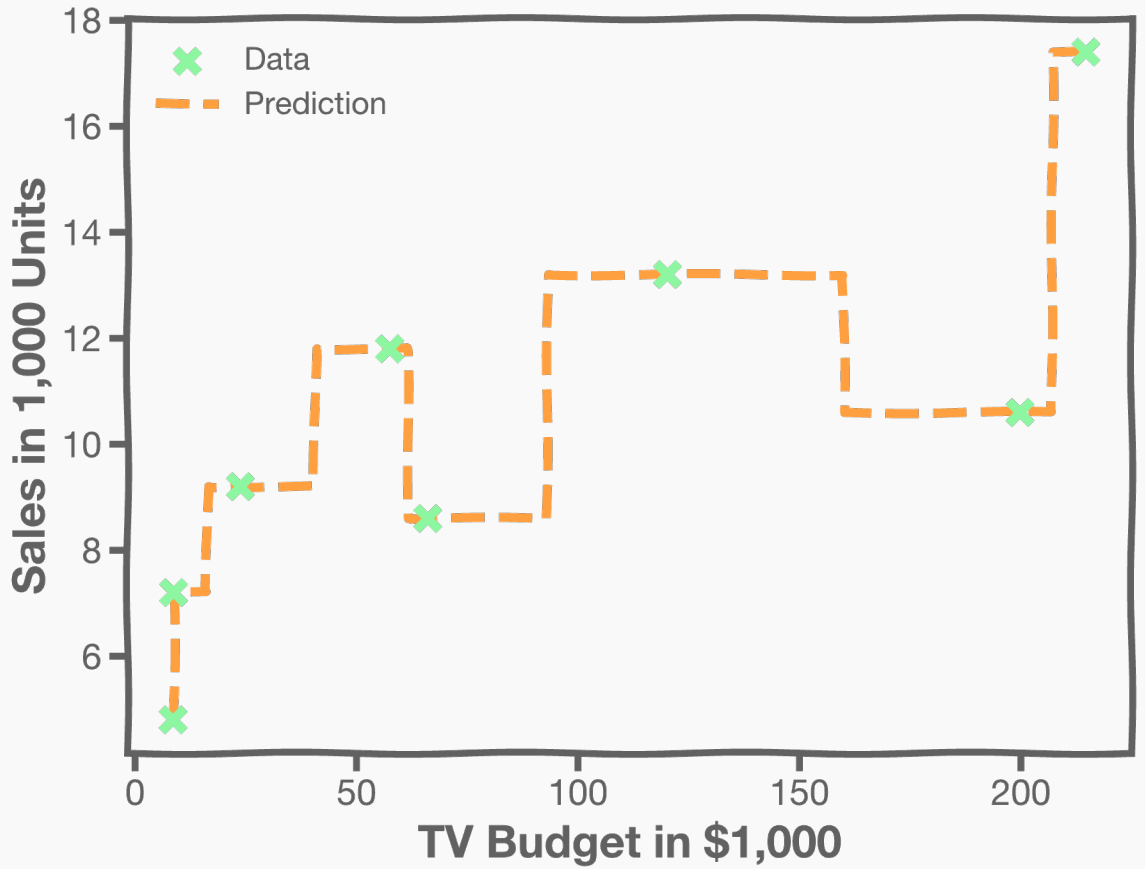
Find distances to all other points $D(x_q, x_i)$

Find the nearest neighbor, (x_p, y_p)

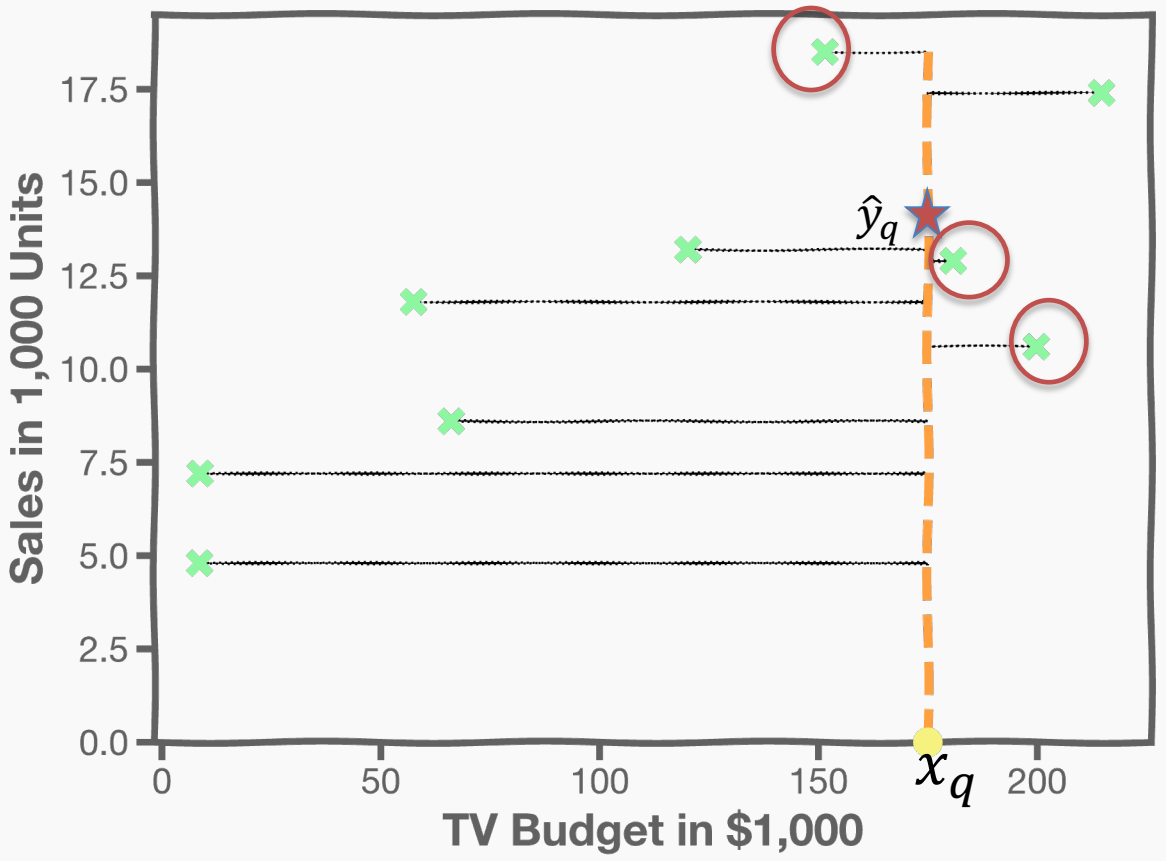
Predict $\hat{y}_q = y_p$

k-Nearest Neighbors – kNN

Do the same for "all" $x's$



k-Nearest Neighbors – kNN



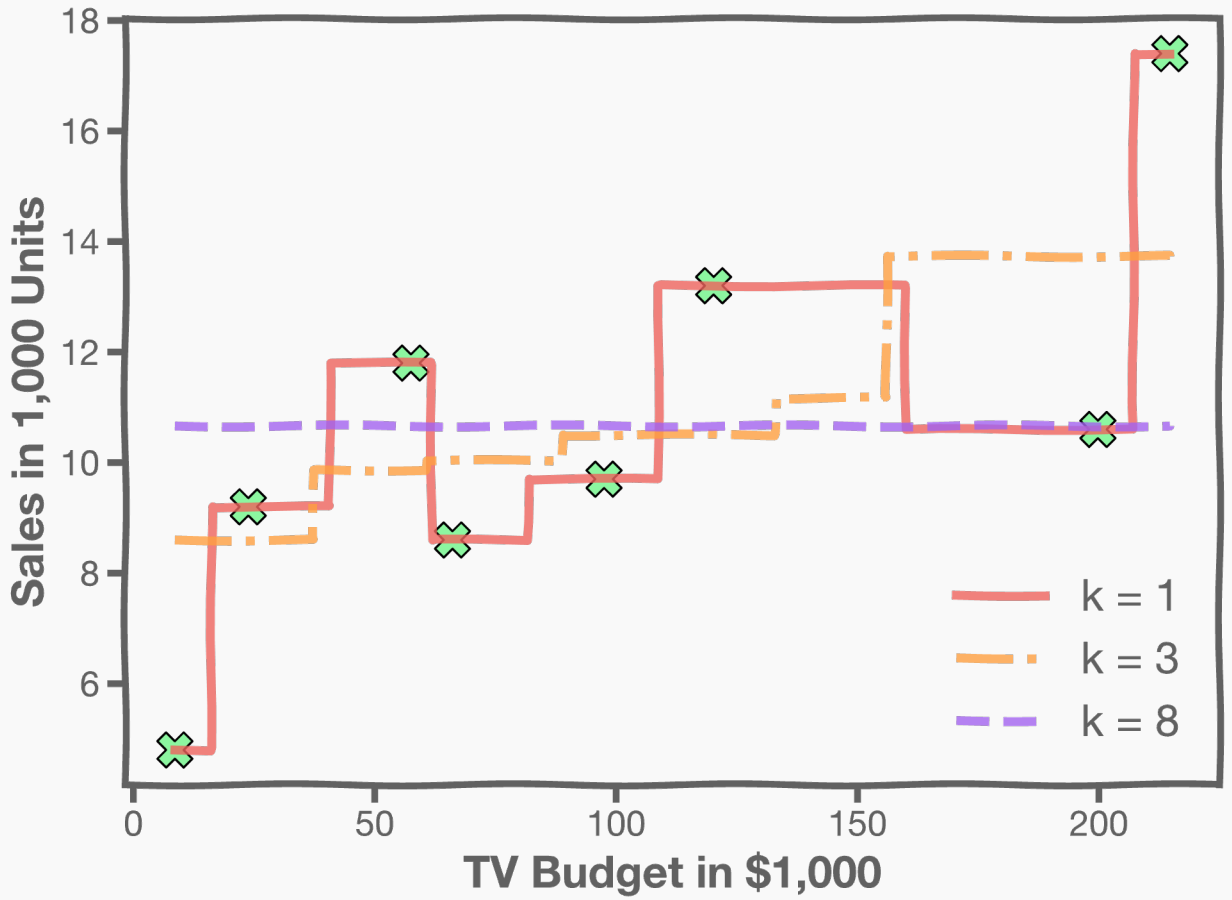
What is $\hat{y}_q$ at some x_q ?

Find distances to all other points $D(x_q, x_i)$

Find the k-nearest neighbors, $x_{q_1}, \ldots, x_{q_k}$

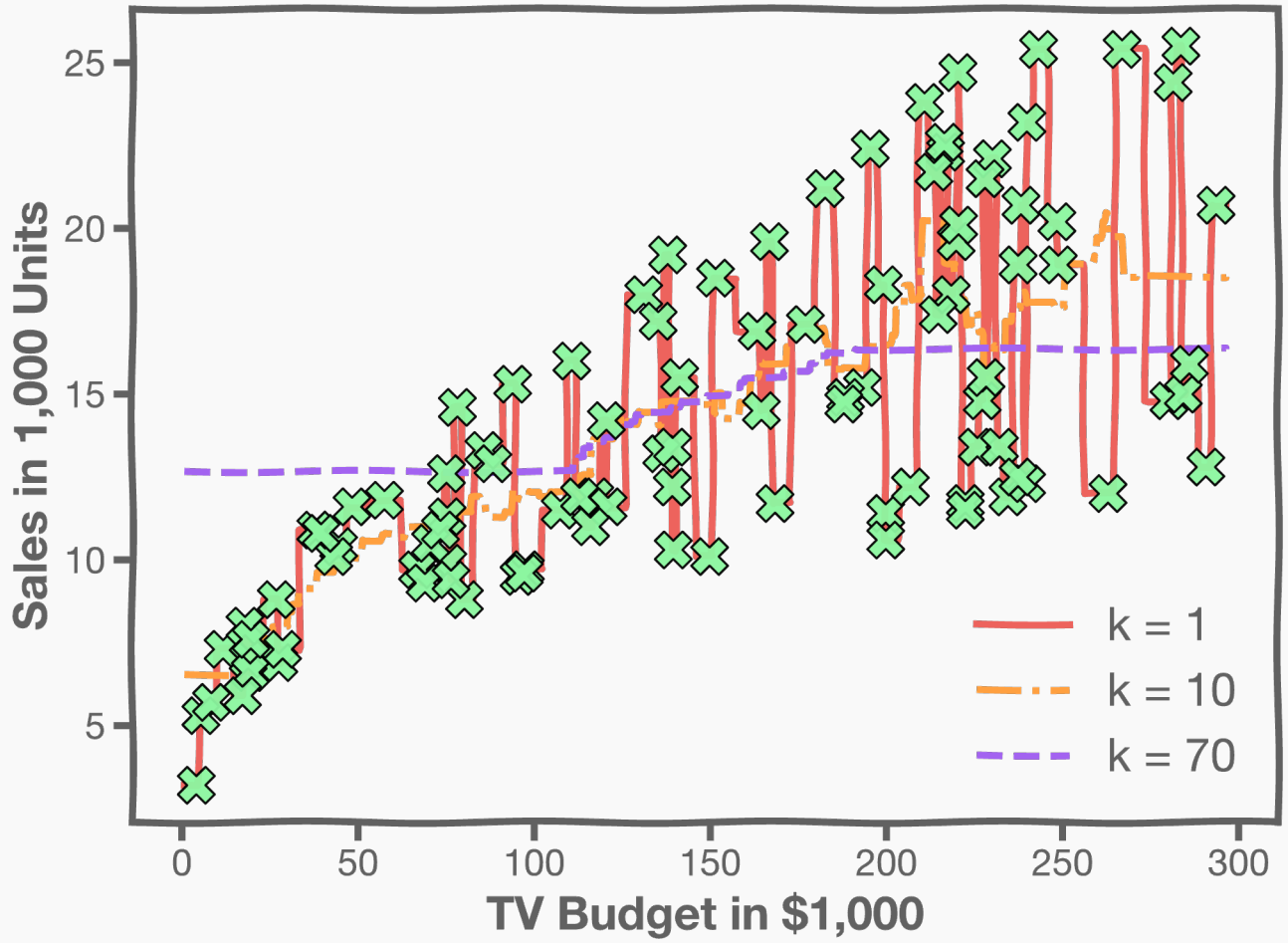
Predict $\hat{y}_q = \frac{1}{k}\sum_i^k y_{q_i}$

k-Nearest Neighbors – kNN



k-Nearest Neighbors – kNN

We can try different k-models on more data



k-Nearest Neighbors – kNN

The very human way of decision making by similar examples.

kNN is a **non-parametric** learning algorithm.

We call kNN non-parametric because it doesn't assume a fixed form for the relationship between input and output.

In simpler words, we only rely on data to get the prediction.

We will be looking at examples where we do assume a fixed form in the future lectures.



k-Nearest Neighbors – kNN

The very human way of decision making by similar examples.
kNN is a **non-parametric** learning algorithm.

The k-Nearest Neighbor Algorithm:

Given a dataset $D = \{(x^{(1)}, y^{(1)}), \dots, (x^{(N)}, y^{(N)})\}$, for every new X:

1. Find the k-number of observations in D most similar to X:

$$\{(x^{(n_1)}, y^{(n_1)}), \dots, (x^{(n_k)}, y^{(n_k)})\}$$

   These are called the k-nearest neighbors of x

2. Average the output of the k-nearest neighbors of x

$$\hat{y} = \frac{1}{K}\sum_{k=1}^{K} y^{(n_k)}$$

k-Nearest Neighbors – kNN

$$\hat{y} = \frac{1}{K}\sum_{k=1}^{K} y^{(n_k)}$$

Remember:
- **Non-parametric doesn't mean 'no hyperparameters'!**
- **It means the model doesn't learn fixed parameters from the data.**

K here is what we call a hyper-parameter.

It is a parameter that we choose before the model training.

We will learn more about these hyper-parameters in the upcoming lectures!